9.3 Then for the heat equation

$$u_t = c^2 u_{xx} \quad \text{on an infinite bar } (-\infty < x < \infty)$$

we can take the Fourier transform on both sides:

$$\mathcal{F}(u_t) = \mathcal{F}(c^2 u_{xx})$$

$$\Rightarrow \quad \frac{\partial \hat{u}}{\partial t} = -c^2 \omega^2 \hat{u}$$

We may now integrate both sides:

$$\int_{0}^{t} \frac{1}{\hat{u}} \frac{\partial \hat{u}}{\partial s}\, ds \Big|_{\omega = \text{const}} = \int_{0}^{t} -c^2 \omega^2 \, ds \Big|_{\omega = \text{const}}$$

$$\Rightarrow \quad \ln(\hat{u}) = -c^2 \omega^2 t + h(\omega)$$

$$\Rightarrow \quad \hat{u} = P(\omega)\, e^{-c^2 \omega^2 t}$$

9.4 Since $u(x,0) = f(x)$

then

$$\mathcal{F}(u(x,0)) = \mathcal{F}(f(x))$$

$$\Rightarrow \hat{u}(\omega, 0) = \hat{f}(\omega)$$

So from $\hat{u}(\omega, t) = P(\omega) e^{-c^2\omega^2 t}$

we get: $P(\omega) = \hat{f}(\omega)$

In other words: $\hat{u}(\omega, t) = \hat{f}(\omega) e^{-c^2\omega^2 t}$

9.5 We may now solve for $u(x,t)$ by simply using the inverse Fourier transform:

$$u(x,t) = \mathcal{F}^{-1}\left(\hat{u}(\omega,t)\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-c^2\omega^2 t} \cdot e^{i\omega x} \, d\omega$$

9.6 Now, suppose for $g(x)$, $\mathcal{F}\{g(x)\} = \hat{g}(\omega) = \frac{1}{\sqrt{2\pi}} e^{-c^2\omega^2 t}$

Then $u(x,t) = \displaystyle\int_{-\infty}^{\infty} \hat{f}(\omega)\hat{g}(\omega) e^{i\omega x} \, d\omega$

Then, we get the (known) result:

$$u(x,t) = \underbrace{(f * g)}_{\text{convolution}}(x) = \int_{-\infty}^{\infty} f(p) g(x-p) \, dp \quad (*)$$

9.7: We also have: $g(x) = \mathcal{F}^{-1}\left(\hat{g}(\omega)\right) = \mathcal{F}^{-1}\left[\frac{1}{\sqrt{2\pi}} e^{-c^2\omega^2 t}\right]$

$$\Rightarrow g(x) = \frac{1}{\sqrt{2c^2 t}\,\sqrt{2\pi}} e^{-x^2/(4c^2 t)}.$$

Replacing x with $x - p$ and substituting this into (~~18~~ *) we finally have

$$u(x,t) = (f * g)(x) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(p) \exp\left\{-\frac{(x-p)^2}{4c^2 t}\right\} dp. \tag{**}$$